# Topic 4: Solved Problems

Step-by-step solutions of problems 1-8 of the official sheet on phase diagrams (Problems T4_PhaseDiagrams I), with the tie line and the lever rule of Session 7, written in four phases and checked against the official solutions.

**DATA READ ON THE DIAGRAMS OF THE PROBLEM SHEET**

| System | Eutectic | α limit | β limit | Melting points |
|---|---|---|---|---|
| Pb-Sn (sheet p. 1) | 61.9 wt% Sn, 183 °C | 19.2 wt% Sn | 97.5 wt% Sn | 327 °C (Pb), 232 °C (Sn) |
| Ag-Cu (sheet p. 2) | 28.1 wt% Cu, 779.1 °C | 8.8 wt% Cu | 92 wt% Cu | 961.93 °C (Ag), 1084.87 °C (Cu) |

`Problem 1` `Problems T4_PhaseDiagrams I.pdf #1` `Lever rule · two equations`

## Phase Limits from the Phase Fractions of Two Alloys

`Original statement · Problems T4_PhaseDiagrams I.pdf`

1. *In the phase diagram of two hypothetic metals, A and B, there is a phase α rich in A and a phase β rich in B. Given the mass fraction of two alloys for a specific temperature (table), determine the composition of the phase limits (or solid solubility limits) of the α and β phases at such temperature.*

| *Alloy composition* | *Fraction of α phase* | *Fraction of β phase* |
|---|---|---|
| *30% B* | *0.78* | *0.22* |
| *65% B* | *0.36* | *0.64* |

**Official Solution:** $C_\alpha = 11.7\%$, $C_\beta = 95\%$

`PHASE 1 · HYPOTHESES & PARAMETERS`

### Hypotheses

1. **Equilibrium:** both alloys are at the same temperature, in equilibrium, and both contain α and β only, so both lie in the α + β two-phase region [Session 7 Slide 26].
2. **One tie line:** at a fixed temperature and pressure, a two-component, two-phase system has $F = C - P + 1 = 2 - 2 + 1 = 1$ degree of freedom, already used by the temperature [Session 7 Slide 13]. The compositions of α and β are therefore the same for both alloys: $C_\alpha$ and $C_\beta$ are the two ends of one tie line.
3. **Mass fractions** obey the lever rule [Session 7 Slide 16].

**Data**

| Quantity | Alloy 1 | Alloy 2 |
|---|---|---|
| Overall composition $C_0$ (% B) | 30 | 65 |
| $W_\alpha$ | 0.78 | 0.36 |
| $W_\beta$ | 0.22 | 0.64 |

**Unknowns:** $C_\alpha$ and $C_\beta$ (% B). Two unknowns and two independent equations (one per alloy; $W_\beta = 1 - W_\alpha$ adds nothing new).

PHASE 2 · FRAMES & STRATEGY

The only frame is the phase diagram itself: composition in % B along the horizontal axis and temperature along the vertical axis. All the work is done on one horizontal line (the tie line at the given temperature), so it is one-dimensional and no change of basis is involved.

**Why the lever rule.** It is the tool that relates the amounts of the phases to the positions of the alloy and of the phase limits on the tie line [Session 7 Slide 16]:

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha}$$

Writing it once for each alloy gives two linear equations in $C_\alpha$ and $C_\beta$.

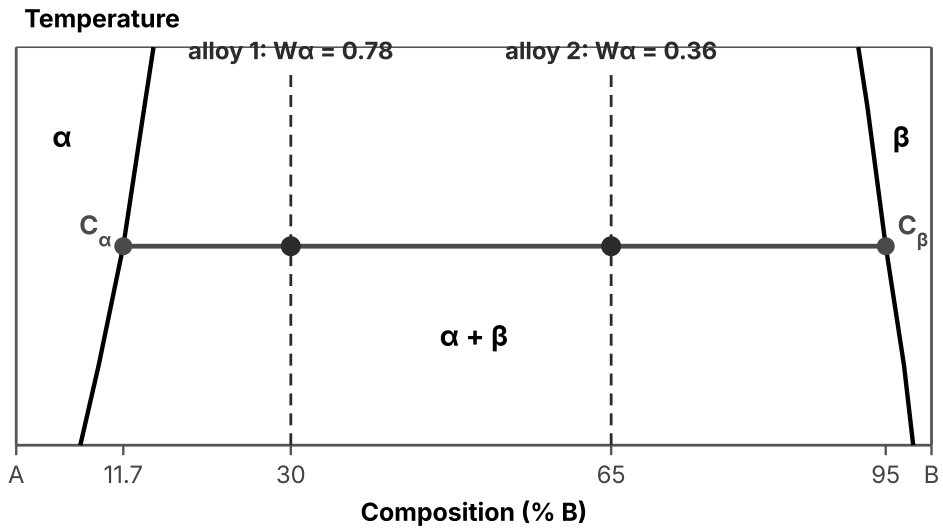



Figure 1. Both alloys are in the α + β region at the same temperature, so they share the tie line from C_α to C_β; the two lever-rule equations give C_α = 11.7 % B and C_β = 95 % B.

PHASE 3 · STEP-BY-STEP DERIVATION

**1. The two equations** Call $D = C_\beta - C_\alpha$ the length of the tie line. Multiplying the lever rule by $D$:

$$W_\alpha D = C_\beta - C_0$$

Alloy 1 ($C_0 = 30$, $W_\alpha = 0.78$) and alloy 2 ($C_0 = 65$, $W_\alpha = 0.36$):

$$0.78\,D = C_\beta - 30 \qquad (1)$$

$$0.36\,D = C_\beta - 65 \qquad (2)$$

**2. Length of the tie line** Subtracting (2) from (1) eliminates $C_\beta$:

$$0.78\,D - 0.36\,D = (C_\beta - 30) - (C_\beta - 65) \;\Rightarrow\; 0.42\,D = 35 \;\Rightarrow\; D = \frac{35}{0.42} = 83.33\ \%\mathrm{B}$$

**3. Limit of β** From (1):

$$C_\beta = 30 + 0.78 \times 83.33 = 30 + 65.00 = 95.0\ \%\mathrm{B}$$

**4. Limit of α**

$$C_\alpha = C_\beta - D = 95.0 - 83.33 = 11.67 \approx 11.7\ \%\mathrm{B}$$

PHASE 4 · CHECKS & INTERPRETATION

**Checks**

- **β fractions** (not used in the solution): $W_\beta = \dfrac{C_0 - C_\alpha}{C_\beta - C_\alpha}$ gives $\dfrac{30 - 11.67}{83.33} = 0.22$ and $\dfrac{65 - 11.67}{83.33} = 0.64$, exactly the values of the table.
- **Mass balance of B** for alloy 1: $0.78 \times 11.67 + 0.22 \times 95.0 = 9.10 + 20.90 = 30.0\ \%\text{B} = C_0$.
- **Order:** $C_\alpha = 11.7 < 30 < 65 < 95 = C_\beta$, so both alloys are indeed inside the α + β region, as assumed.
- **Units:** all compositions are in % B (mass), fractions are dimensionless; each equation is homogeneous in % B.

**Interpretation** At this temperature α can dissolve at most 11.7 % B and β at most 5 % A (95 % B). The alloy closer to the α limit (30 % B) has more α, as the lever rule predicts.

**Check against the official solution**

The official solution gives $C_\alpha = 11.7\%$ and $C_\beta = 95\%$; the result agrees. The official solutions PDF has no worked page for this problem; only the key of the statement is available.

Problem 2 | Problems T4_PhaseDiagrams I.pdf #2 | Pb-Sn · hypereutectic

# Primary and Total β in a Pb-Sn Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf

*1. From 2.8 kg of a Sn-Pb alloy, is it possible to have 2.21 kg of primary β and 2.53 kg of β total at (183 – ΔT) °C? Justify your answer.*

**Official Solution: Yes, for an alloy with composition $C_0 = 90\%$ B**

PHASE 1 · HYPOTHESES & PARAMETERS

### Hypotheses

1. **Equilibrium cooling** with the Pb-Sn diagram of the problem sheet (Problems T4_PhaseDiagrams I.pdf, p. 1): eutectic at $C_e = 61.9$ wt% Sn and 183 °C; at 183 °C the α limit is $C_\alpha = 19.2$ wt% Sn and the β limit $C_\beta = 97.5$ wt% Sn.
2. **Primary β** exists only in a **hypereutectic** alloy ($61.9 < C_0 < 97.5$ wt% Sn): it is the β that forms in the β + L region before the eutectic reaction [Session 7 Slide 31].
3. Just below the eutectic temperature the primary β keeps the amount it had just above it, and the liquid present just above becomes the eutectic [Session 7 Slide 32].
4. "% B" in the key means wt% Sn (B is the Sn-rich side of the diagram).

### Data

| Quantity | Value |
|---|---|
| Mass of alloy $m$ | 2.8 kg |
| Mass of primary β $m_{\beta'}$ | 2.21 kg |
| Mass of total β $m_\beta$ (at 183 – ΔT) | 2.53 kg |
| $C_\alpha$, $C_e$, $C_\beta$ at 183 °C | 19.2, 61.9, 97.5 wt% Sn |

**Unknown:** the composition $C_0$. There is one unknown and two conditions, so the question is whether **both** masses can be met by the same $C_0$.

PHASE 2 · FRAMES & STRATEGY

The frame is the Pb-Sn diagram, composition in wt% Sn and temperature in °C; the work is on the horizontal tie lines just above and just below 183 °C, so it is one-dimensional and no change of basis is needed.

**Strategy.** The primary β fixes $C_0$ through the lever rule on the short tie line between the eutectic liquid and β (microconstituents). The total β is then **predicted** with the lever rule on the long tie line between α and β (phases), and compared with 2.53 kg. If they agree, the answer is yes.

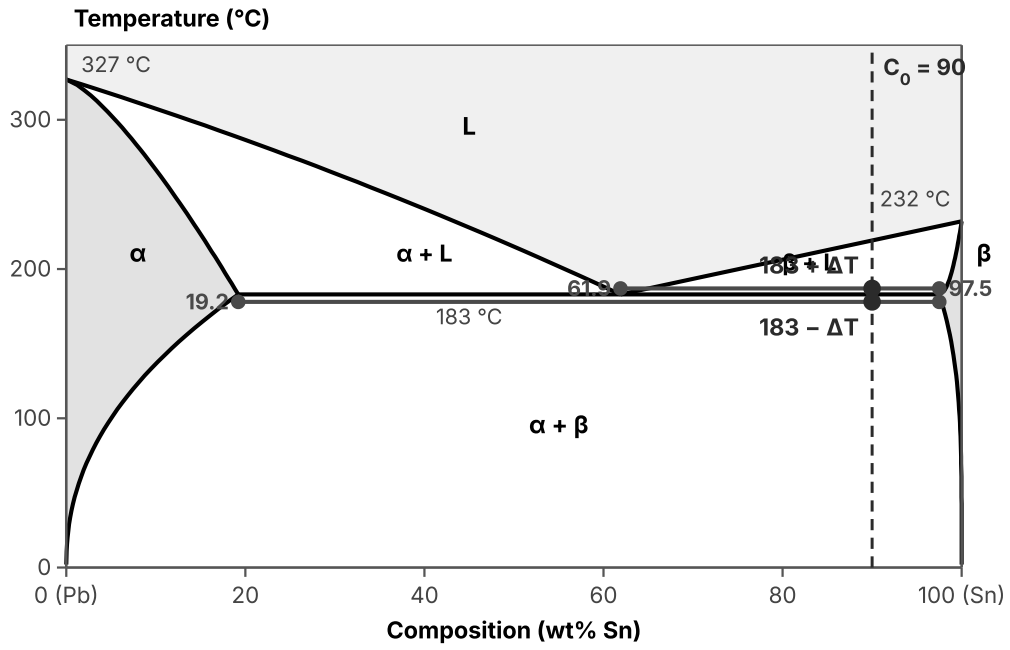



Figure 2. Pb-Sn diagram of the problem sheet (eutectic 61.9 wt% Sn at 183 °C, α 19.2 and β 97.5 wt% Sn). The alloy with 79 % primary β is the hypereutectic Pb-90 wt% Sn; the tie lines just above and just below 183 °C give the primary β and the total β.

PHASE 3 · STEP-BY-STEP DERIVATION

**1. Mass fraction of primary β**

$$W_{\beta'} = \frac{m_{\beta'}}{m} = \frac{2.21\ \text{kg}}{2.8\ \text{kg}} = 0.7893$$

**2. Composition of the alloy** Just above 183 °C the alloy is β (97.5 wt% Sn) plus liquid of eutectic composition (61.9 wt% Sn). The lever rule gives the β fraction, which is the primary β [Session 7 Slide 32]:

$$W_{\beta'} = \frac{C_0 - C_e}{C_\beta - C_e} = \frac{C_0 - 61.9}{97.5 - 61.9} = \frac{C_0 - 61.9}{35.6}$$

Solving for $C_0$:

$$C_0 = 61.9 + 0.7893 \times 35.6 = 61.9 + 28.10 = 90.0 \text{ wt\% Sn}$$

$61.9 < 90.0 < 97.5$: the alloy is hypereutectic, as primary β requires.

**3. Total β predicted for this alloy** Just below 183 °C the phases are α (19.2 wt% Sn) and β (97.5 wt% Sn):

$$W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{90.0 - 19.2}{97.5 - 19.2} = \frac{70.8}{78.3} = 0.9042$$

$$m_\beta = W_\beta \, m = 0.9042 \times 2.8 \text{ kg} = 2.532 \text{ kg} \approx 2.53 \text{ kg}$$

**4. Answer** Both conditions are met by the same alloy: **yes**, it is possible, with a Pb-90 wt% Sn alloy.

PHASE 4 · CHECKS & INTERPRETATION

**Checks**

- **Eutectic part:** mass of eutectic $= 2.8 - 2.21 = 0.59$ kg; β inside it $= 2.53 - 2.21 = 0.32$ kg, a fraction $0.32/0.59 = 0.54$. The eutectic of the sheet diagram has $W_{\beta,e} = (61.9 - 19.2)/(97.5 - 19.2) = 42.7/78.3 = 0.545$ of β, and $0.545 \times 0.59 = 0.32$ kg: consistent.
- **Units:** masses in kg, fractions dimensionless, compositions in wt% Sn.

**Interpretation** The two masses are not independent: once the primary β fixes the composition, the total β is fixed too. A different total β (say 2.40 kg) would have been impossible for any alloy.

**Check against the official solution**

The official key ("Yes, for an alloy with composition $C_0 = 90\%$") agrees. The worked page (Solution problems Phase diagrams I.pdf, p. 6) finds $C_0 = 90\%$ from the primary β only; the check of the total β (2.53 kg) above completes the justification. The page writes the fraction as "079"; it is 0.79.

`Problem 3` `Problems T4_PhaseDiagrams I.pdf #3` `Pb-Sn · hypereutectic`

# Phases and Microconstituents of a Pb-80 wt% Sn Alloy

`Original statement · Problems T4_PhaseDiagrams I.pdf`

1. *Using the Pb-Sn phase diagram for an alloy composition of Pb-80%Sn, indicate at the temperatures of $T_1 = 250$ ºC, $T_2 = 183 + \Delta T$ ºC, $T_3 = 183 - \Delta T$ ºC, $T_4 = 100$ ºC: a) The phases present, chemical composition of the phases and mass fraction. b) The micro constituents, chemical composition and mass fractions. Note the agreement that exists between the data obtained in (a) and (b).*

**Official Solution: a)** $T_1$: $C_L = 80\%$ **Sn,** $w_L = 1$; $T_2$: $C_L = 61.9\%$ **Sn,** $C_\beta = 97.5\%$ **Sn,** $w_L = 0.49$; $T_3$: $C_\alpha = 19.2\%$ **Sn,** $C_\beta = 97.5\%$ **Sn,** $w_\alpha = 0.22$; $T_4$: $C_\alpha = 5\%$ **Sn,** $C_\beta = 100\%$ **Sn,** $w_\alpha = 0.21$; **b)** $C_{\beta'} = 97.5\%$ **Sn,** $C_E = 61.9\%$ **Sn,** $w_E = 0.49$, $w_{\beta'} = 0.51$

`PHASE 1 · HYPOTHESES & PARAMETERS`

### Hypotheses

1. **Equilibrium cooling**, Pb-Sn diagram of the problem sheet (p. 1): eutectic at 61.9 wt% Sn and 183 °C; at 183 °C, α of 19.2 and β of 97.5 wt% Sn.
2. Values at 250 °C and 100 °C are **readings of the figure**: at 250 °C the alloy is above the liquidus on the Sn side (which goes from 183 °C at 61.9 wt% Sn to 232 °C at 100 wt% Sn); at 100 °C the solvus lines give α of about 5 and β of about 100 wt% Sn.
3. $C_0 = 80$ wt% Sn lies between 61.9 and 97.5: a **hypereutectic** alloy, so the primary phase is β [Session 7 Slide 31].

### Data

| Temperature | Region | Phase compositions (wt% Sn) |
|---|---|---|
| $T_1 = 250$ °C | L | $C_L = 80$ |
| $T_2 = 183 + \Delta T$ | β + L | $C_L = 61.9$, $C_\beta = 97.5$ |
| $T_3 = 183 - \Delta T$ | α + β | $C_\alpha = 19.2$, $C_\beta = 97.5$ |
| $T_4 = 100$ °C | α + β | $C_\alpha = 5$, $C_\beta = 100$ |

PHASE 2 · FRAMES & STRATEGY

The frame is the Pb-Sn diagram (wt% Sn, °C). At each temperature the work is on one horizontal tie line, so it is one-dimensional; no change of basis is needed.

**Method.** (a) Tie line for the compositions, lever rule for the fractions [Session 7 Slides 15-16]. (b) Below the eutectic temperature, the lever rule on the short tie line (β to eutectic liquid) gives the microconstituents, and the long tie line (α to β) gives the phases [Session 7 Slide 32]; comparing both answers part (b)'s "agreement".

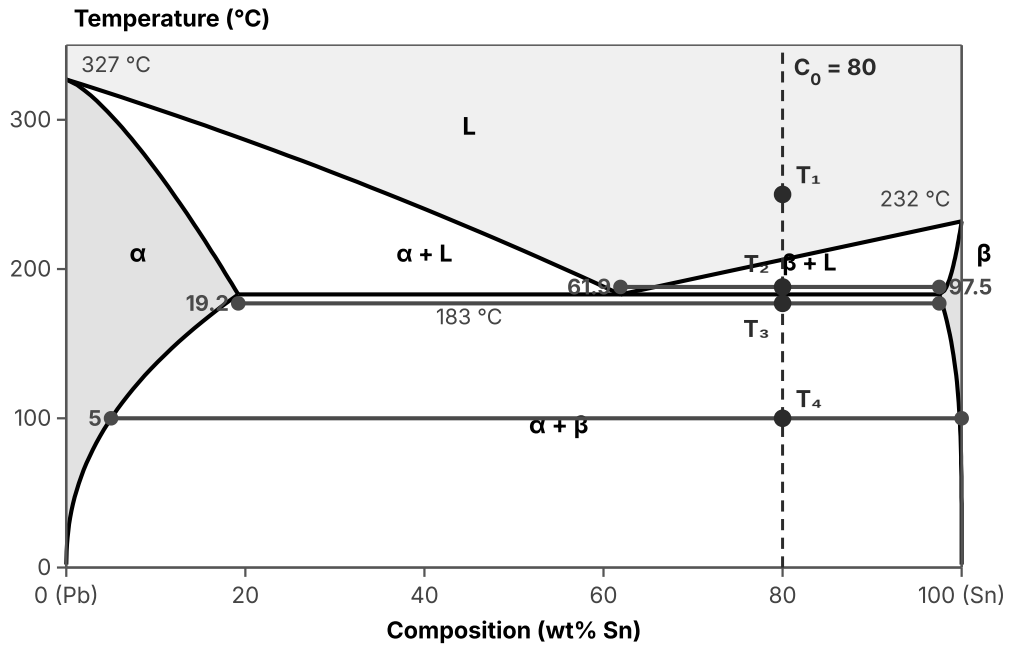



Figure 3. Pb-80 wt% Sn at $T_1$ = 250 °C (liquid), $T_2$ = 183 + ΔT (β + L), $T_3$ = 183 − ΔT (α + β) and $T_4$ = 100 °C (α + β, with α of 5 and β of 100 wt% Sn, a reading of the figure). Problem sheet diagram.

PHASE 3 · STEP-BY-STEP DERIVATION

**a) Phases $T_1 = 250$ °C.** Above the liquidus: one phase, liquid with the composition of the alloy. $C_L = 80$ wt% Sn, $W_L = 1$.

$T_2 = 183 + \Delta T$**.** Region β + L; tie line from $C_L = 61.9$ to $C_\beta = 97.5$ (length $97.5 - 61.9 = 35.6$):

$$W_\beta = \frac{C_0 - C_L}{C_\beta - C_L} = \frac{80 - 61.9}{97.5 - 61.9} = \frac{18.1}{35.6} = 0.508 \qquad W_L = \frac{C_\beta - C_0}{C_\beta - C_L} = \frac{97.5 - 80}{35.6} = \frac{17.5}{35.6} = 0.492$$

$T_3 = 183 - \Delta T$**.** Region α + β; tie line from 19.2 to 97.5 (length 78.3):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{97.5 - 80}{97.5 - 19.2} = \frac{17.5}{78.3} = 0.224 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{80 - 19.2}{78.3} = \frac{60.8}{78.3} = 0.776$$

$T_4 = 100$ **°C.** Region α + β; tie line from 5 to 100 (length 95):

$$W_\alpha = \frac{100 - 80}{100 - 5} = \frac{20}{95} = 0.211 \qquad W_\beta = \frac{80 - 5}{95} = \frac{75}{95} = 0.789$$

**b) Microconstituents** At $T_1$ and $T_2$ the microconstituents are the phases themselves (liquid; primary β + liquid).

At $T_3$ there are two microconstituents: **primary β** (97.5 wt% Sn), formed in the β + L region, and **eutectic** (overall 61.9 wt% Sn), formed from the liquid present at $T_2$. With $P = C_0 - C_e$ and $Q = C_\beta - C_0$ on the short tie line:

$$W_{\beta'} = \frac{C_0 - C_e}{C_\beta - C_e} = \frac{80 - 61.9}{97.5 - 61.9} = 0.508 = W_\beta(T_2) \qquad W_E = \frac{C_\beta - C_0}{C_\beta - C_e} = \frac{17.5}{35.6} = 0.492 = W_L(T_2)$$

**Phases inside the eutectic:**

$$W_{\beta,E} = W_\beta(T_3) - W_{\beta'} = 0.776 - 0.508 = 0.268 \qquad W_{\alpha,E} = W_\alpha(T_3) = 0.224$$

(all the α is in the eutectic, since the primary phase is β).

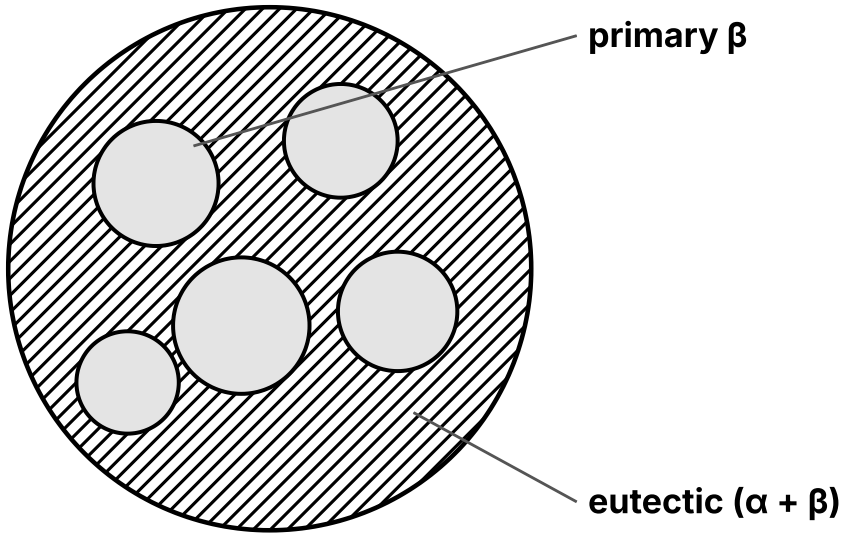



Figure 4. Pb-80 wt% Sn below the eutectic temperature: primary β (51 %) in the eutectic α + β (49 %).

PHASE 4 · CHECKS & INTERPRETATION

**Agreement between (a) and (b)**

- $W_{\beta,E} + W_{\alpha,E} = 0.268 + 0.224 = 0.492 = W_E$: the phases of the eutectic add up to the eutectic.
- Inside the eutectic, α is $0.224/0.492 = 0.455$, the same as the eutectic product of the sheet diagram, $(97.5 - 61.9)/(97.5 - 19.2) = 35.6/78.3 = 0.455$.
- Mass balance at $T_3$: $0.224 \times 19.2 + 0.776 \times 97.5 = 4.30 + 75.66 = 80.0$ wt% Sn.

**Interpretation** Below 183 °C the alloy is about half primary β and half eutectic. On further cooling to 100 °C the solvus lines move outwards (α purer in Pb, β practically pure Sn) and the amounts of the phases change slightly (α from 0.224 to 0.211) by solid-state precipitation, while the microconstituents keep their shape.

**Check against the official solution**

All the values agree with the key ($w_L = 0.49$, $w_\alpha = 0.22$, $w_\alpha = 0.21$; $w_E = 0.49$, $w_{\beta'} = 0.51$).

ERRATUM IN THE WORKED SOLUTION

Solution problems Phase diagrams I.pdf, p. 7, writes "T2; phases α and L"; the phases at $T_2$ are **β and L** (the alloy is hypereutectic). On the same page the $T_3$ formula is printed with the denominator $97.5 - 61.9$, but the value 0.22 corresponds to $97.5 - 19.2 = 78.3$: $17.5/35.6$ would give 0.49, while $17.5/78.3 = 0.22$. The numbers of the page are right; the labels and that denominator are not.

Problem 4 | Problems T4_PhaseDiagrams I.pdf #4 | Pb-Sn · hypoeutectic

# Phases and Microconstituents of a Pb-30 wt% Sn Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf

1. *Using the Pb-Sn phase diagram for an alloy composition of Pb-30%Sn, indicate at the temperatures of $T_1 = 250$ ºC, $T_2 = 183 + \Delta T$ ºC, $T_3 = 183 - \Delta T$ ºC, $T_4 = 100$ ºC: a) The phases present, chemical composition of the phases and mass fraction. b) The micro constituents, chemical composition and mass fractions. Note the agreement that exists between the data obtained in (a) and (b).*

**Official Solution: a)** $T_1$: $C_L = 36\%$ **Sn,** $C_\alpha = 12\%$ **Sn,** $w_L = 0.75$; $T_2$: $C_L = 61.9\%$ **Sn,** $C_\alpha = 19.2\%$ **Sn,** $w_L = 0.25$; $T_3$: $C_\alpha = 19.2\%$ **Sn,** $C_\beta = 97.5\%$ **Sn,** $w_\alpha = 0.86$; $T_4$: $C_\alpha = 5\%$ **Sn,** $C_\beta = 100\%$ **Sn,** $w_\alpha = 0.74$; **b)** $T_3$: $C_{\alpha'} = 19.2\%$ **Sn,** $C_E = 61.9\%$ **Sn,** $w_E = 0.25$, $w_{\alpha'} = 0.75$

PHASE 1 · HYPOTHESES & PARAMETERS

### Hypotheses

1. **Equilibrium cooling**, Pb-Sn diagram of the problem sheet (p. 1): eutectic at 61.9 wt% Sn and 183 °C; α of 19.2 and β of 97.5 wt% Sn at 183 °C.
2. **Readings of the figure:** at 250 °C the tie line meets the α solidus at 12 wt% Sn and the liquidus at 36 wt% Sn; at 100 °C the solvus lines give α of 5 and β of 100 wt% Sn (the same readings as the official solution).
3. $19.2 < C_0 = 30 < 61.9$: a **hypoeutectic** alloy, so the primary phase is α [Session 7 Slide 30]. This is the same kind of alloy as the hypoeutectic example of the slides.

### Data

| Temperature | Region | Phase compositions (wt% Sn) |
|---|---|---|
| $T_1 = 250$ °C | α + L | $C_\alpha = 12$, $C_L = 36$ |
| $T_2 = 183 + \Delta T$ | α + L | $C_\alpha = 19.2$, $C_L = 61.9$ |
| $T_3 = 183 - \Delta T$ | α + β | $C_\alpha = 19.2$, $C_\beta = 97.5$ |
| $T_4 = 100$ °C | α + β | $C_\alpha = 5$, $C_\beta = 100$ |

PHASE 2 · FRAMES & STRATEGY

Frame: the Pb-Sn diagram (wt% Sn, °C); one horizontal tie line per temperature, so no change of basis is needed.

**Method.** Tie line and lever rule at each temperature [Session 7 Slides 15-16]; below 183 °C, the short tie line (α to eutectic liquid) gives the microconstituents and the long one (α to β) the phases [Session 7 Slide 32].

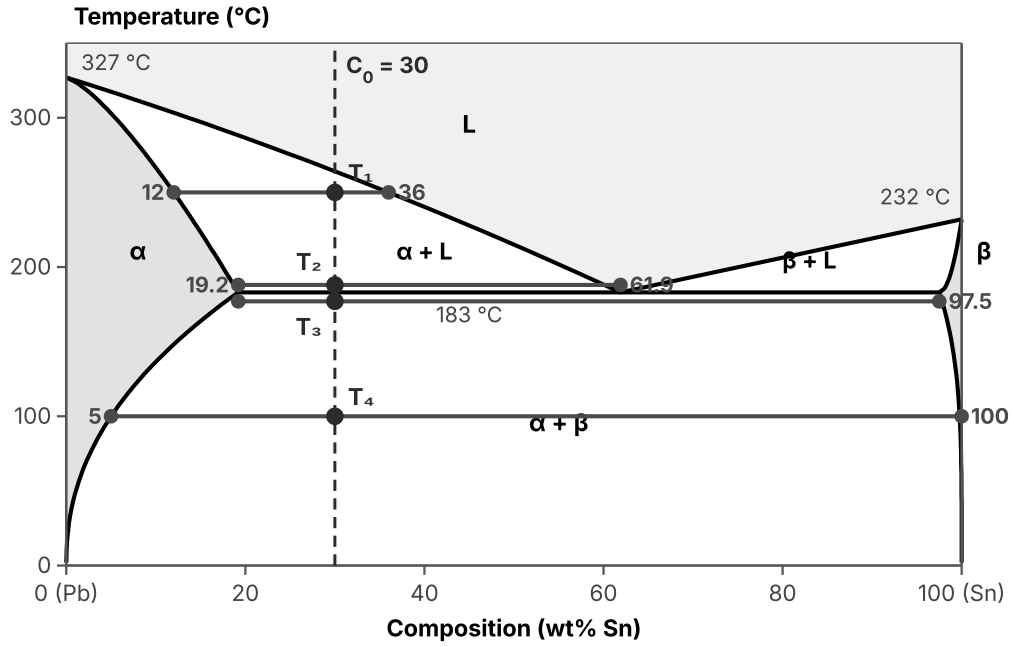



Figure 5. Pb-30 wt% Sn at $T_1$ = 250 °C (α of 12 and L of 36 wt% Sn, a reading of the figure), $T_2$ = 183 + ΔT (α + L), $T_3$ = 183 − ΔT (α + β) and $T_4$ = 100 °C. Problem sheet diagram.

PHASE 3 · STEP-BY-STEP DERIVATION

**a) Phases $T_1 = 250$ °C.** α + L; tie line from 12 to 36 (length 24):

$$W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{30 - 12}{36 - 12} = \frac{18}{24} = 0.750 \qquad W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{36 - 30}{24} = \frac{6}{24} = 0.250$$

**$T_2 = 183 + \Delta T$.** α + L; tie line from 19.2 to 61.9 (length 42.7):

$$W_L = \frac{30 - 19.2}{61.9 - 19.2} = \frac{10.8}{42.7} = 0.253 \qquad W_\alpha = \frac{61.9 - 30}{42.7} = \frac{31.9}{42.7} = 0.747$$

$T_3 = 183 - \Delta T$**.** α + β; tie line from 19.2 to 97.5 (length 78.3):

$$W_\alpha = \frac{97.5 - 30}{97.5 - 19.2} = \frac{67.5}{78.3} = 0.862 \qquad W_\beta = \frac{30 - 19.2}{78.3} = \frac{10.8}{78.3} = 0.138$$

$T_4 = 100$ **°C.** α + β; tie line from 5 to 100 (length 95):

$$W_\alpha = \frac{100 - 30}{100 - 5} = \frac{70}{95} = 0.737 \qquad W_\beta = \frac{30 - 5}{95} = \frac{25}{95} = 0.263$$

**b) Microconstituents** At $T_1$ and $T_2$ they coincide with the phases (α + liquid). At $T_3$: **primary α** (19.2 wt% Sn) and **eutectic** (61.9 wt% Sn overall):

$$W_{\alpha'} = \frac{C_e - C_0}{C_e - C_\alpha} = \frac{61.9 - 30}{61.9 - 19.2} = 0.747 = W_\alpha(T_2) \qquad W_E = \frac{C_0 - C_\alpha}{C_e - C_\alpha} = \frac{30 - 19.2}{42.7} = 0.253 = W_L(T_2)$$

**Phases inside the eutectic:**

$$W_{\alpha,E} = W_\alpha(T_3) - W_{\alpha'} = 0.862 - 0.747 = 0.115 \qquad W_{\beta,E} = W_\beta(T_3) = 0.138$$

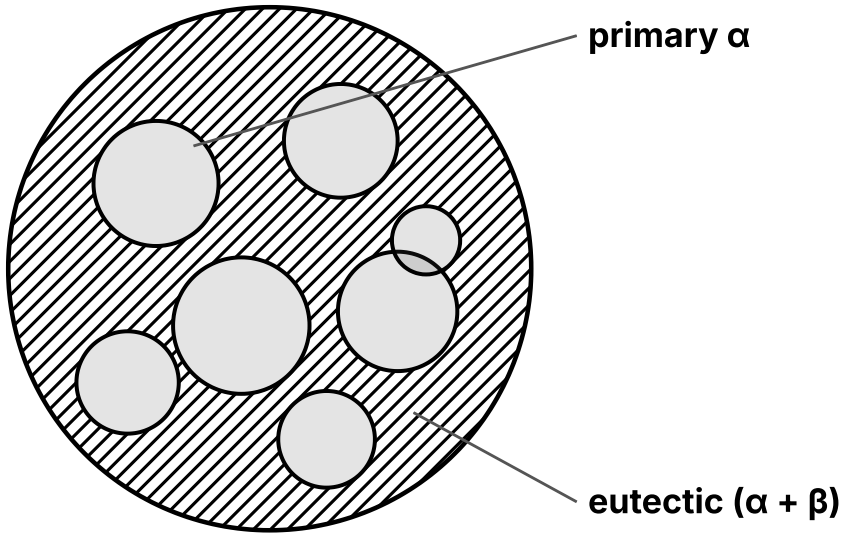



Figure 6. Pb-30 wt% Sn below the eutectic temperature: primary α (75 %) in the eutectic α + β (25 %).

PHASE 4 · CHECKS & INTERPRETATION

**Agreement between (a) and (b)**

- $W_{\alpha,E} + W_{\beta,E} = 0.115 + 0.138 = 0.253 = W_E$.
- Inside the eutectic, α is $0.115/0.253 = 0.455$, the eutectic product of the sheet diagram ( $35.6/78.3 = 0.455$), the same as in Problem 3: the eutectic has the same make-up whatever the alloy.
- Mass balance at $T_3$: $0.862 \times 19.2 + 0.138 \times 97.5 = 16.55 + 13.45 = 30.0$ wt% Sn.

**Interpretation** The alloy is three quarters primary α in one quarter of eutectic, the structure of the hypoeutectic micrograph of Slide 30. From 250 °C to 183 °C the amount of α grows from 25 % to 75 % while the liquid gets richer in Sn, from 36 to 61.9 wt%.

**Check against the official solution**

The values agree with the key ($w_L = 0.75, 0.25$; $w_\alpha = 0.86, 0.74$; $w_E = 0.25$, $w_{\alpha'} = 0.75$). The α inside the eutectic is $0.862 - 0.747 = 0.115$; the worked page writes 0.11 because it subtracts the rounded values $0.86 - 0.75$.

ERRATUM IN THE WORKED SOLUTION

Solution problems Phase diagrams I.pdf, p. 10, gives for $T_4$ the numbers of Problem 3: $\chi_\alpha = (100 - 80)/(100 - 5) = 0.21$ and $\chi_{\beta,\text{total}} = 0.79$. For this alloy ($C_0 = 30$) the correct values are $W_\alpha = 70/95 = 0.74$ and $W_\beta = 0.26$, as the key of the statement says. The $T_1$ formula on p. 9 is also printed with the wrong labels ($C_o - C_\beta$ over $C_E - C_\alpha$), while its numbers, $(30 - 12)/(36 - 12) = 0.75$, are right.

Problem 5 | Problems T4_PhaseDiagrams I.pdf #5 | Ag-Cu · eutectic

# The Eutectic Ag-28.1 wt% Cu Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 2

*1. Given an alloy Ag-28.1%Cu, indicate for the given temperatures, $T_1 = 900$ °C; $T_2 = (780 + \Delta T)$ °C; $T_3 = (780 - \Delta T)$ °C: a. Present phases, chemical composition of phases and mass fraction. b. Microconstituents, chemical composition and mass fraction at $T_3$.*

**Official Solution: a) $T_1$: Liquid (28.1%Cu); $T_2$: Liquid (28.1%Cu); $T_3$: $\alpha$($C_\alpha$ = 8.8%Cu) + $\beta$($C_\beta$ = 92%Cu); $x_\alpha = 0.77$; $x_\beta = 0.23$. b) Eutectic microconstituent (28.1%Cu); $C_{\alpha E}$ = 8.8%Cu; $C_{\beta E}$ = 92%Cu**

PHASE 1 · HYPOTHESES & PARAMETERS

### Hypotheses

1. **Equilibrium cooling** with the Ag-Cu diagram of the problem sheet (p. 2): eutectic at 28.1 wt% Cu and 779.1 °C (the statement rounds it to 780 °C); at that temperature (Ag) = α contains 8.8 wt% Cu and (Cu) = β contains 92 wt% Cu. Melting points: 961.93 °C (Ag) and 1084.87 °C (Cu).
2. $C_0 = 28.1$ wt% Cu is exactly the **eutectic composition** [Session 7 Slides 26-27].

### Data

| Quantity | Value |
|---|---|
| $C_0$ | 28.1 wt% Cu |
| $C_\alpha$, $C_e$, $C_\beta$ at the eutectic temperature | 8.8, 28.1, 92 wt% Cu |
| Eutectic temperature | 779.1 °C ≈ 780 °C |

PHASE 2 · FRAMES & STRATEGY

Frame: the Ag-Cu diagram (wt% Cu, °C), one horizontal tie line per temperature; no change of basis is needed.

**Why no lever rule above 780 °C.** The eutectic liquid meets both liquidus lines at their lowest point, so it never enters a two-phase L + solid region: it stays liquid down to the eutectic temperature and then transforms at once, $L \rightarrow \alpha + \beta$ [Session 7 Slide 27].

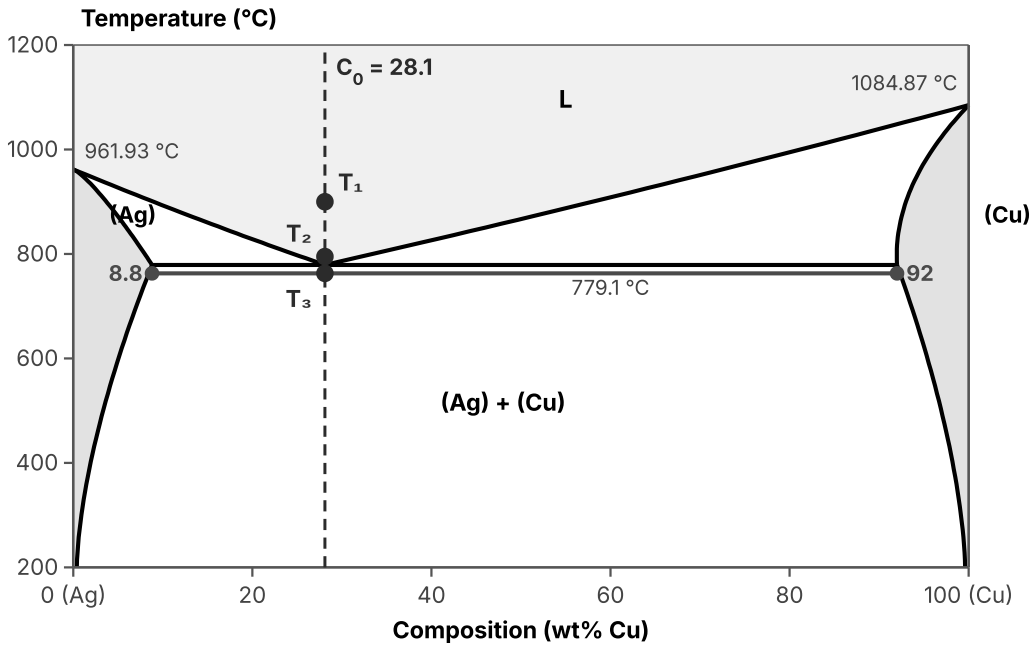



Figure 7. Ag-Cu diagram of the problem sheet. The eutectic alloy is liquid down to 779.1 °C and then becomes (Ag) of 8.8 and (Cu) of 92 wt% Cu.

PHASE 3 · STEP-BY-STEP DERIVATION

**a) Phases $T_1 = 900$ °C.** Above the liquidus: liquid, $C_L = 28.1$ wt% Cu, $W_L = 1$.

$T_2 = 780 + \Delta T$**.** Still above the eutectic point: liquid, $C_L = 28.1$ wt% Cu, $W_L = 1$.

$T_3 = 780 - \Delta T$**.** α + β; tie line from 8.8 to 92 (length $92 - 8.8 = 83.2$):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{92 - 28.1}{92 - 8.8} = \frac{63.9}{83.2} = 0.768 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{28.1 - 8.8}{83.2} = \frac{19.3}{83.2} = 0.232$$

**b) Microconstituents at $T_3$** All the liquid of $T_2$ became eutectic: **one microconstituent, the eutectic**, $W_E = 1$, with overall composition 28.1 wt% Cu. Inside it: α (8.8 wt% Cu) with mass fraction 0.768 and β (92 wt% Cu) with mass fraction 0.232.

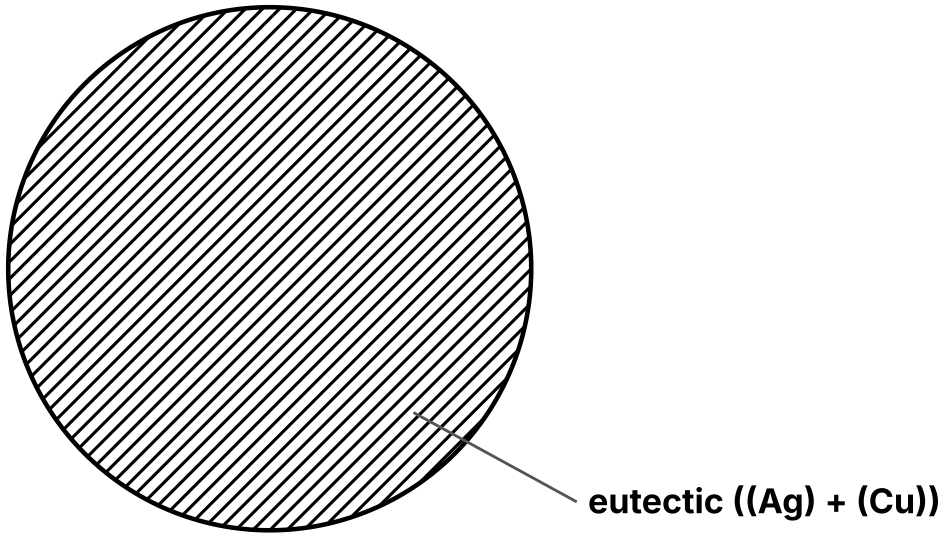



Figure 8. Ag-28.1 wt% Cu below 779 °C: 100 % eutectic, (Ag) and (Cu) in alternating layers.

PHASE 4 · CHECKS & INTERPRETATION

**Checks**

- $W_\alpha + W_\beta = 0.768 + 0.232 = 1$.
- Mass balance: $0.768 \times 8.8 + 0.232 \times 92 = 6.76 + 21.34 = 28.1$ wt% Cu $= C_0$.
- Phase rule at the eutectic point (1 atm): $F = 2 - 3 + 1 = 0$, so the transformation happens at one temperature, with a plateau in the cooling curve, like a pure metal [Session 7 Slide 27].

**Interpretation** The eutectic alloy has the lowest melting point of the system (779 °C, well below 961.93 °C for Ag and 1084.87 °C for Cu) and solidifies at a single temperature into a fine layered mixture of a Ag-rich and a Cu-rich solid.

**Check against the official solution**

Agrees with the key: $x_\alpha = 0.77$, $x_\beta = 0.23$; 100 % eutectic. The official solutions PDF has no worked page for this problem.

Problem 6 | Problems T4_PhaseDiagrams I.pdf #6 | Ag-Cu · hypereutectic

# The Hypereutectic Ag-80 wt% Cu Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 2

1. *Using the Cu-Ag phase diagram, for an alloy with composition $C_0$ = 80% Cu, for temperatures $T_1 = 900$ °C; $T_2 = (780 + \Delta T)$ °C; $T_3 = (780 - \Delta T)$ °C, determine: a. Present phases, chemical composition of phases and mass fraction. b. Microconstituents, chemical composition and mass fraction at $T_3$.*

**Official Solution: a)** $T_1$: $C_L$ = 58%Cu; $C_\beta$ = 93%Cu; $X_L = 0.37$; $X_\beta = 0.63$; $T_2$: $C_L$ = 28.1%Cu; $C_\beta$ = 92%Cu; $X_L = 0.19$; $X_\beta = 0.81$; $T_3$: $C_\alpha$ = 8.8%Cu; $C_\beta$ = 92%Cu; $X_\alpha = 0.14$; $X_\beta = 0.86$. **b) eutectic microconstituent ($X_E = 0.19$) and proeutectic β ($X_{\beta p} = 0.81$)**

PHASE 1 · HYPOTHESES & PARAMETERS

### Hypotheses

1. **Equilibrium cooling**, Ag-Cu diagram of the problem sheet (p. 2): eutectic 28.1 wt% Cu at 779.1 °C ≈ 780 °C; α = (Ag) of 8.8 and β = (Cu) of 92 wt% Cu at that temperature.
2. **Reading of the figure at 900 °C:** the tie line meets the Cu-side liquidus at 58 wt% Cu and the (Cu) solidus at 93 wt% Cu (the readings of the official solution).
3. $28.1 < C_0 = 80 < 92$: a **hypereutectic** alloy, primary phase β [Session 7 Slide 31].

### Data

| Temperature | Region | Phase compositions (wt% Cu) |
|---|---|---|
| $T_1 = 900$ °C | L + β | $C_L = 58$, $C_\beta = 93$ |
| $T_2 = 780 + \Delta T$ | L + β | $C_L = 28.1$, $C_\beta = 92$ |
| $T_3 = 780 - \Delta T$ | α + β | $C_\alpha = 8.8$, $C_\beta = 92$ |

PHASE 2 · FRAMES & STRATEGY

Frame: the Ag-Cu diagram (wt% Cu, °C), one tie line per temperature; no change of basis. Lever rule for phases [Session 7 Slide 16]; short tie line (eutectic liquid to β) for the microconstituents [Session 7 Slide 32].

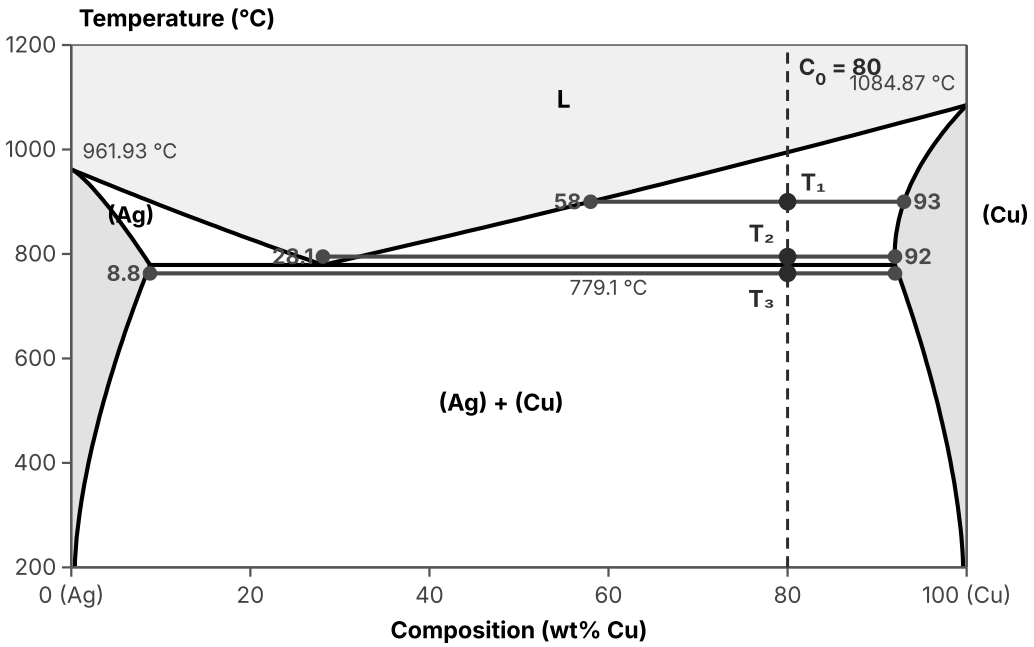



Figure 9. Ag-80 wt% Cu at $T_1$ = 900 °C (L of 58 and (Cu) of 93 wt% Cu, a reading of the figure), $T_2$ = 780 + ΔT and $T_3$ = 780 − ΔT. Problem sheet diagram.

PHASE 3 · STEP-BY-STEP DERIVATION

**a) Phases** $T_1 = 900$ **°C** (tie line length $93 - 58 = 35$):

$$W_L = \frac{C_\beta - C_0}{C_\beta - C_L} = \frac{93 - 80}{93 - 58} = \frac{13}{35} = 0.371 \qquad W_\beta = \frac{C_0 - C_L}{C_\beta - C_L} = \frac{80 - 58}{35} = \frac{22}{35} = 0.629$$

$T_2 = 780 + \Delta T$ (length $92 - 28.1 = 63.9$):

$$W_L = \frac{92 - 80}{92 - 28.1} = \frac{12}{63.9} = 0.188 \qquad W_\beta = \frac{80 - 28.1}{63.9} = \frac{51.9}{63.9} = 0.812$$

$T_3 = 780 - \Delta T$ (length $92 - 8.8 = 83.2$):

$$W_\alpha = \frac{92 - 80}{92 - 8.8} = \frac{12}{83.2} = 0.144 \qquad W_\beta = \frac{80 - 8.8}{83.2} = \frac{71.2}{83.2} = 0.856$$

**b) Microconstituents at $T_3$ Proeutectic (primary) β** of 92 wt% Cu, and **eutectic** of 28.1 wt% Cu overall:

$$W_{\beta p} = \frac{C_0 - C_e}{C_\beta - C_e} = \frac{80 - 28.1}{92 - 28.1} = \frac{51.9}{63.9} = 0.812 \qquad W_E = \frac{C_\beta - C_0}{C_\beta - C_e} = \frac{12}{63.9} = 0.188$$

**Phases inside the eutectic:**

$$W_{\beta,E} = W_\beta(T_3) - W_{\beta p} = 0.856 - 0.812 = 0.044 \qquad W_{\alpha,E} = W_\alpha(T_3) = 0.144$$

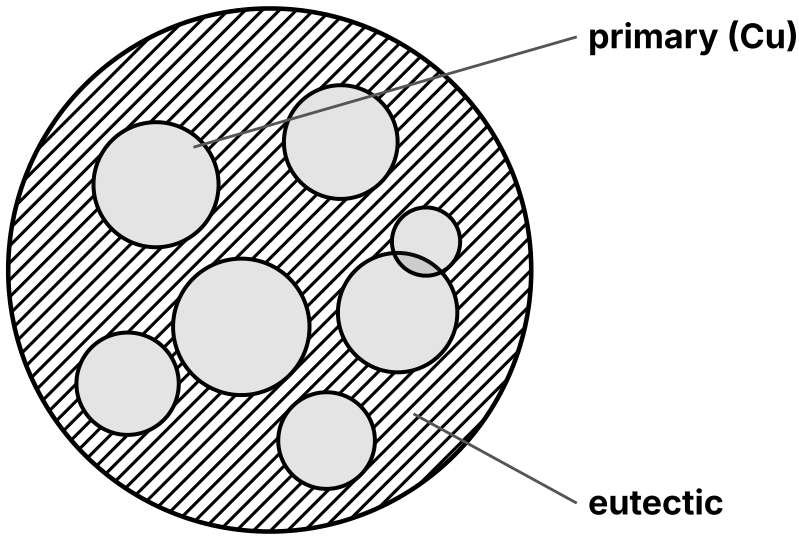



Figure 10. Ag-80 wt% Cu below 779 °C: primary (Cu) (81 %) in the eutectic (19 %).

PHASE 4 · CHECKS & INTERPRETATION

**Checks**

- $W_{\alpha,E} + W_{\beta,E} = 0.144 + 0.044 = 0.188 = W_E$.
- Inside the eutectic α is $0.144/0.188 = 0.768$, the make-up of the eutectic alloy of Problem 5.
- Mass balance at $T_3$: $0.1442 \times 8.8 + 0.8558 \times 92 = 1.27 + 78.73 = 80.0$ wt% Cu.

**Interpretation** Between 900 °C and the eutectic temperature the amount of primary (Cu) grows from 63 % to 81 %, and the liquid becomes poorer in Cu, from 58 to 28.1 wt%. The final structure is mostly primary (Cu) crystals with about 19 % eutectic between them.

**Check against the official solution**

All values agree with the key. The worked page (Solution problems Phase diagrams I.pdf, pp. 11-12) adds a fourth temperature, $T_4 = 500$ °C, that is not in the statement of the sheet, and writes $\chi_{Cu,E} = 0.86 - 0.81 = 0.04$ (exactly $0.856 - 0.812 = 0.044$).

Problem 7 | Problems T4_PhaseDiagrams I.pdf #7 | Ag-Cu · hypoeutectic

# The Hypoeutectic Ag-10 wt% Cu Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 2

1. *Using the Cu-Ag phase diagram, for an alloy with composition $C_0$ = 10% Cu, for the temperatures $T_1 = 900$ °C, $T_2 = (780 + \Delta T)$ °C, $T_3 = (780 - \Delta T)$ °C, determine: a. Present phases, chemical composition of phases and mass fraction. b. Microconstituents, chemical composition and mass fraction at $T_3$.*

**Official Solution: a) $T_1$: Liquid (10%Cu); $T_2$: $C_L$ = 28.1%Cu; $C_\alpha$ = 8.8%Cu; $X_\alpha = 0.94$; $X_L = 0.06$; $T_3$: $C_\alpha$ = 8.8%Cu; $C_\beta$ = 92%Cu; $X_\alpha = 0.98$; $X_\beta = 0.02$. b) eutectic microconstituent ($X_E = 0.06$) and proeutectic α ($X_{\alpha p} = 0.94$)**

PHASE 1 · HYPOTHESES & PARAMETERS

### Hypotheses

1. **Equilibrium cooling**, Ag-Cu diagram of the problem sheet (p. 2): eutectic 28.1 wt% Cu at 779.1 °C ≈ 780 °C; α = (Ag) of 8.8 and β = (Cu) of 92 wt% Cu.
2. **Reading of the figure at 900 °C:** the Ag-side liquidus falls from 961.93 °C (0 % Cu) to 779.1 °C (28.1 % Cu), and at 10 wt% Cu it is just below 900 °C, so at 900 °C the alloy is still liquid (as the official solution states).
3. $8.8 < C_0 = 10 < 28.1$: a **hypoeutectic** alloy, very close to the α limit; the primary phase is α [Session 7 Slide 30].

### Data

| Temperature | Region | Phase compositions (wt% Cu) |
|---|---|---|
| $T_1 = 900$ °C | L | $C_L = 10$ |
| $T_2 = 780 + \Delta T$ | α + L | $C_\alpha = 8.8$, $C_L = 28.1$ |
| $T_3 = 780 - \Delta T$ | α + β | $C_\alpha = 8.8$, $C_\beta = 92$ |

PHASE 2 · FRAMES & STRATEGY

Frame: the Ag-Cu diagram (wt% Cu, °C), one tie line per temperature; no change of basis. Lever rule for the phases [Session 7 Slide 16] and for the microconstituents (short tie line from α to the eutectic liquid) [Session 7 Slide 32].

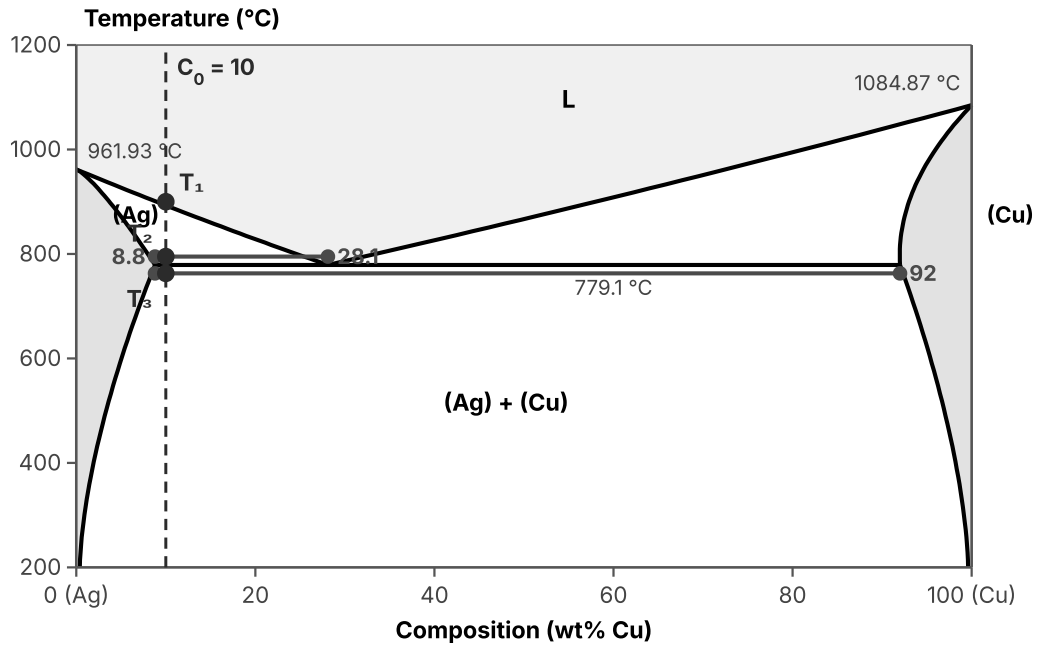



Figure 11. Ag-10 wt% Cu at $T_1$ = 900 °C (liquid, just above the liquidus, a reading of the figure), $T_2$ = 780 + ΔT and $T_3$ = 780 − ΔT. Problem sheet diagram.

PHASE 3 · STEP-BY-STEP DERIVATION

**a) Phases $T_1 = 900$ °C.** Liquid, $C_L = 10$ wt% Cu, $W_L = 1$.

$T_2 = 780 + \Delta T$ (tie line length $28.1 - 8.8 = 19.3$):

$$W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{28.1 - 10}{28.1 - 8.8} = \frac{18.1}{19.3} = 0.938 \qquad W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{10 - 8.8}{19.3} = \frac{1.2}{19.3} = 0.062$$

$T_3 = 780 - \Delta T$ (length $92 - 8.8 = 83.2$):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{92 - 10}{92 - 8.8} = \frac{82}{83.2} = 0.986 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{10 - 8.8}{83.2} = \frac{1.2}{83.2} = 0.014$$

**b) Microconstituents at $T_3$ Proeutectic (primary) α** of 8.8 wt% Cu and **eutectic** of 28.1 wt% Cu overall:

$$W_{\alpha p} = \frac{C_e - C_0}{C_e - C_\alpha} = \frac{18.1}{19.3} = 0.938 \qquad W_E = \frac{C_0 - C_\alpha}{C_e - C_\alpha} = \frac{1.2}{19.3} = 0.062$$

**Phases inside the eutectic:**

$$W_{\alpha,E} = W_\alpha(T_3) - W_{\alpha p} = 0.986 - 0.938 = 0.048 \qquad W_{\beta,E} = W_\beta(T_3) = 0.014$$

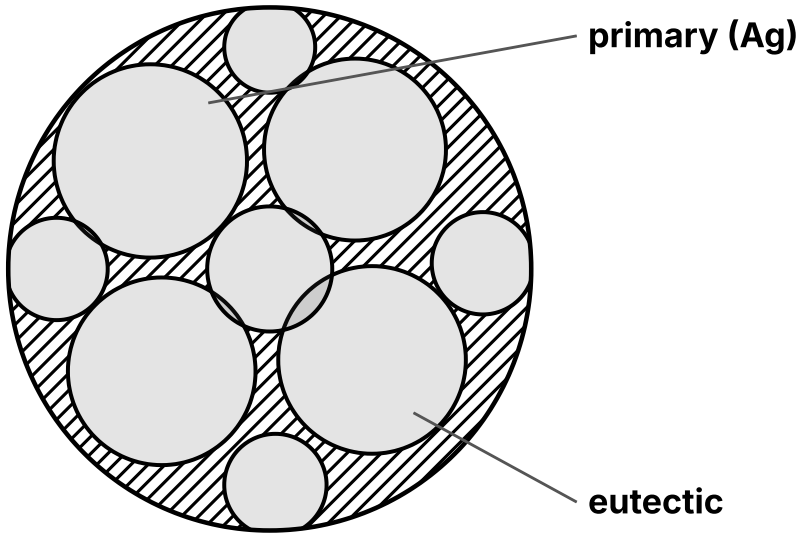



Figure 12. Ag-10 wt% Cu below 779 °C: primary (Ag) (94 %) with a little eutectic (6 %) between the grains.

PHASE 4 · CHECKS & INTERPRETATION

**Checks**

- $W_{\alpha,E} + W_{\beta,E} = 0.048 + 0.014 = 0.062 = W_E$; inside the eutectic α is $0.048/0.062 = 0.77$, as in Problem 5.
- Mass balance at $T_3$: $0.9856 \times 8.8 + 0.0144 \times 92 = 8.67 + 1.33 = 10.0$ wt% Cu.

**Interpretation** The alloy is only 1.2 wt% Cu beyond the solubility limit of α, so almost all of it is primary (Ag) and just 6 % is eutectic, found as thin films between the α grains. An alloy below 8.8 wt% Cu would form no eutectic at all.

**Check against the official solution**

At $T_1$, $T_2$ and in part (b) the results agree with the key.

**DISCREPANCY WITH THE OFFICIAL SOLUTION**

At $T_3$ the key gives $X_\alpha = 0.98$ and $X_\beta = 0.02$. With the data of the problem, $W_\alpha = 82/83.2 = 0.9856$ and $W_\beta = 1.2/83.2 = 0.0144$, which round to 0.99 and 0.01. The computed values are kept; the key's 0.98 and 0.02 are not reproduced by any rounding of the data.

Problem 8 | Problems T4_PhaseDiagrams I.pdf #8 | Diagram construction

# Drawing a Eutectic Diagram and the 40 % B Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 3

1. *Two metals (A and B) have partial solubility among each other. Metal A, with a melting point at 700 °C, can dissolve 5% of B at 0 °C, and its maximum solubility occurs at 400 °C, being 25% B. On the other hand, B metal melts at 600 °C, can dissolve 10% of A at 0 °C and its maximum solubility occurs at 400 °C, being 20% A. At 400 °C there is an eutectic reaction when the B content is 65%. a. Draw and identify the parts of the phase diagram in equilibrium (suppose straight lines). b. For an alloy 40% B, determine the phases composition and their relative proportions at 450 °C, 400 °C + ΔT and 400 °C – ΔT. c. For an alloy 40% B, draw its microstructure indicating the microconstituents and the mass fraction of each one. Additionally, indicate the mass fraction of each of the phases present in such microconstituents.*

**Official Solution: b) 450 °C:** $C_L$ **= 55%B;** $C_\alpha$ **= 20%B;** $X_\alpha = 0.43$**;** $X_L = 0.57$**; 400 °C + ΔT:** $C_L$ **= 65%B;** $C_\alpha$ **= 25%B;** $X_\alpha = 0.62$**;** $X_L = 0.38$**; 400 °C - ΔT:** $C_\beta$ **= 80%B;** $C_\alpha$ **= 25%B;** $X_\alpha = 0.73$ **;** $X_\beta = 0.27$**. c) Eutectic microconstituent and proeutectic α;** $X_{\alpha p} = 0.62$**;** $X_E = 0.38$**; Phases in the eutectic microconstituent:** $X(\alpha_E) = 0.11$**;** $X(\beta_E) = 0.27$**.**

PHASE 1 · HYPOTHESES & PARAMETERS

### Hypotheses

1. **Binary eutectic system** with partial solid solubility: α (rich in A) and β (rich in B) [Session 7 Slide 26].
2. **Straight lines** between the given points, as the statement asks.
3. Compositions in % B (mass); "20 % A" in β means 80 % B, and "10 % A" means 90 % B.

### Points of the diagram

| Point | Composition (% B) | Temperature (°C) | Meaning |
|---|---|---|---|
| Melting point of A | 0 | 700 | top of α solidus and of the A-side liquidus |
| Melting point of B | 100 | 600 | top of β solidus and of the B-side liquidus |
| Eutectic point E | 65 | 400 | lowest point of the liquid |
| Max. solubility of B in α | 25 | 400 | end of the eutectic line, α side |
| Max. solubility of A in β | 80 | 400 | end of the eutectic line, β side |
| α solvus at 0 °C | 5 | 0 | |

| Point | Composition (% B) | Temperature (°C) | Meaning |
|---|---|---|---|
| β solvus at 0 °C | 90 | 0 | |

PHASE 2 · FRAMES & STRATEGY

Frame: composition (% B) on the horizontal axis and temperature (°C) on the vertical one. Each boundary is a straight segment, so a composition on it at a temperature $T$ comes from linear interpolation between its two end points; the tie lines are horizontal. No change of basis is needed.

**Lines of the diagram (part a):**

- liquidus: $(0, 700) \to (65, 400)$ and $(100, 600) \to (65, 400)$;
- solidus: $(0, 700) \to (25, 400)$ for α and $(100, 600) \to (80, 400)$ for β;
- eutectic line: $(25, 400) \to (80, 400)$;
- solvus: $(25, 400) \to (5, 0)$ for α and $(80, 400) \to (90, 0)$ for β.

Regions: L above the liquidus; α + L and β + L between liquidus and solidus; α and β next to the axes, bounded by solidus and solvus; α + β below the eutectic line, between the solvus lines.

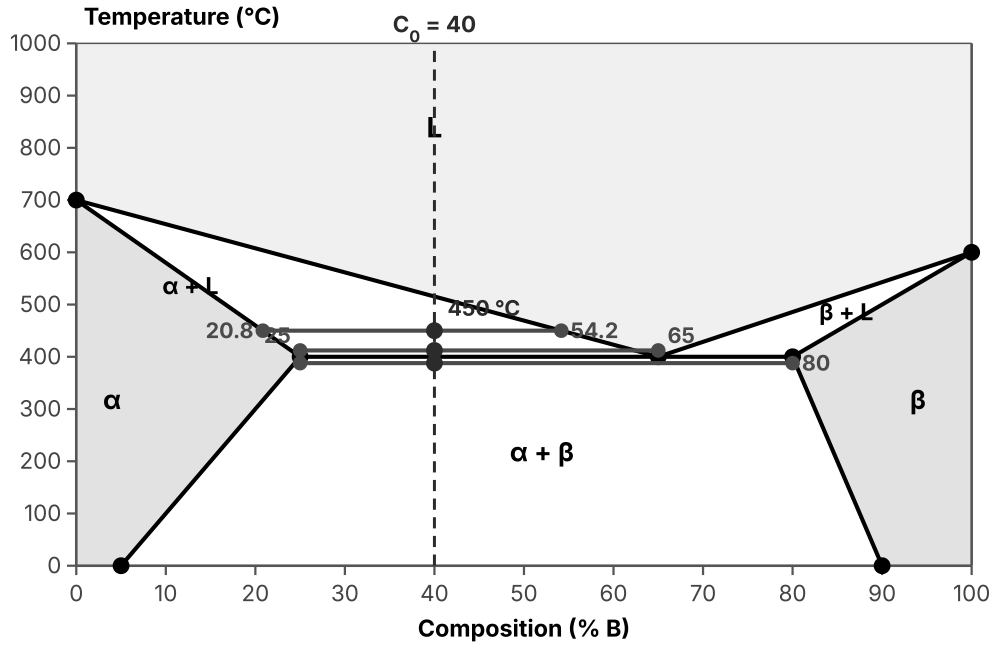



Figure 13. A-B diagram with straight lines: melting points 700 °C (A) and 600 °C (B), eutectic at 400 °C and 65 % B, maximum solubilities 25 % B in α and 20 % A in β (80 % B) at 400 °C, and 5 % B in α and 10 % A in β (90 % B) at 0 °C. Dashed: the 40 % B alloy, with its tie lines at 450 °C and just above and below 400 °C.

PHASE 3 · STEP-BY-STEP DERIVATION

**b) Phases of the 40 % B alloy 450 °C.** The alloy is between the A-side liquidus and the α solidus: **α + L.** On a straight segment from $(0, 700)$ to $(C_1, 400)$ the composition at $T$ is $C = C_1 \dfrac{700 - T}{700 - 400}$:

$$C_L = 65 \times \frac{700 - 450}{300} = 65 \times \frac{250}{300} = 54.17\ \%\text{B} \qquad C_\alpha = 25 \times \frac{250}{300} = 20.83\ \%\text{B}$$

$$W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{40 - 20.83}{54.17 - 20.83} = \frac{19.17}{33.33} = 0.575 \qquad W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{54.17 - 40}{33.33} = \frac{14.17}{33.33} = 0.425$$

**400 °C + ΔT.** Still α + L, with the ends of the tie line at the eutectic temperature: $C_\alpha = 25$, $C_L = 65$ % B (length 40):

$$W_L = \frac{40 - 25}{65 - 25} = \frac{15}{40} = 0.375 \qquad W_\alpha = \frac{65 - 40}{40} = \frac{25}{40} = 0.625$$

**400 °C – ΔT.** α + β; $C_\alpha = 25$, $C_\beta = 80$ % B (length 55):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{80 - 40}{80 - 25} = \frac{40}{55} = 0.727 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{40 - 25}{55} = \frac{15}{55} = 0.273$$

**c) Microconstituents at 400 °C – ΔT** $25 < 40 < 65$: hypoeutectic, so the microconstituents are **proeutectic α** (25 % B) and **eutectic** (65 % B overall). The α present just above 400 °C is the proeutectic α, and the liquid becomes eutectic [Session 7 Slide 32]:

$$W_{\alpha p} = \frac{C_e - C_0}{C_e - C_\alpha} = \frac{65 - 40}{65 - 25} = 0.625 \qquad W_E = \frac{C_0 - C_\alpha}{C_e - C_\alpha} = \frac{40 - 25}{40} = 0.375$$

**Phases inside the eutectic:**

$$W_{\alpha,E} = W_\alpha - W_{\alpha p} = 0.7273 - 0.6250 = 0.1023 \qquad W_{\beta,E} = W_\beta = 0.273$$

The same numbers from the make-up of the eutectic, which is $(80-65)/(80-25) = 15/55 = 0.273$ α and $40/55 = 0.727$ β: $W_{\alpha,E} = 0.375 \times 0.273 = 0.102$ and $W_{\beta,E} = 0.375 \times 0.727 = 0.273$.

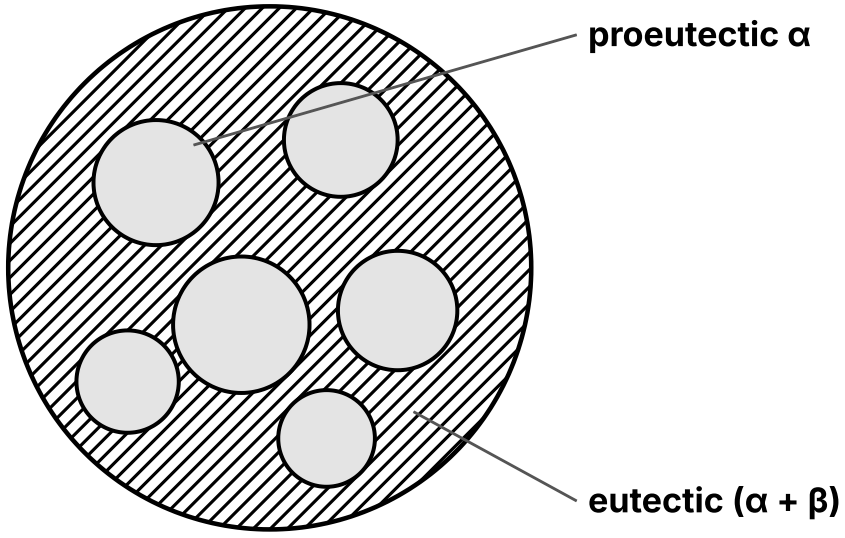



Figure 14. 40 % B alloy at 400 − ΔT: proeutectic α (62.5 %) in the eutectic α + β (37.5 %).

PHASE 4 · CHECKS & INTERPRETATION

**Checks**

- Mass balance at 450 °C: $0.425 \times 20.83 + 0.575 \times 54.17 = 8.85 + 31.15 = 40.0$ % B.
- Mass balance at 400 °C − ΔT: $0.727 \times 25 + 0.273 \times 80 = 18.18 + 21.82 = 40.0$ % B.
- $W_{\alpha,E} + W_{\beta,E} = 0.102 + 0.273 = 0.375 = W_E$, and $W_{\alpha p} + W_{\alpha,E} = 0.625 + 0.102 = 0.727 = W_\alpha$.

**Interpretation** On cooling from the liquid, α starts to form at the liquidus ($700 - 300 \times 40/65 = 515$ °C for this alloy), grows to 62.5 % at the eutectic temperature, and the remaining 37.5 % of liquid, of 65 % B, becomes eutectic at 400 °C. Most of the eutectic is β (0.273 of the alloy), which is all the β of the alloy.

**Check against the official solution**

At 400 °C ± ΔT the results agree with the key (0.62/0.38 are 0.625/0.375; 0.73/0.27). The key writes "$C_\alpha$ = 25%Cu" at 400 °C + ΔT, a typo for 25 % B.

DISCREPANCY WITH THE OFFICIAL SOLUTION

**450 °C.** The key reads $C_L = 55$ and $C_\alpha = 20$ % B from its drawing and gives $X_L = 0.57$, $X_\alpha = 0.43$. With the straight lines that the statement requires, linear interpolation gives $C_L = 54.17$ and $C_\alpha = 20.83$ % B, hence $W_L = 0.575$ and $W_\alpha = 0.425$. The difference comes from reading the drawing; the interpolated values are kept.

**α in the eutectic.** The key gives $X(\alpha_E) = 0.11$, which is $0.73 - 0.62$ with rounded values. Exactly, $0.7273 - 0.6250 = 0.1023$, confirmed by $0.375 \times 15/55 = 0.102$; the key's $0.11$ is a rounding effect.